INCH POUND

A-A-58026A
29 April 2004
Superceding
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April 11, 2002

COMMERCIAL ITEM DESCRIPTION

TRACTOR, AIRCRAFT TOWING, U-30

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers the requirements for one type of U-30 aircraft towing tractor, hereafter referred to as tow tractor. The tow tractor is capable of towing military aircraft up to 850,000 pounds on dry level concrete. The primary intended use of this tow tractor is to tow C-5 military aircraft on US Air Force installations.

2. SALIENT CHARACTERISTICS. Failure of the tow tractor to meet the salient characteristics of this CID shall be cause for rejection of the tow tractor. The tow tractor shall be free from faulty construction and any defect that affects appearance, serviceability, and durability. The tow tractor shall be representative of the quality and workmanship found in tow tractors offered for new commercial sale.

2.1 Gross vehicle weight (GVW). The tow tractor shall have a minimum GVW of 100,000 pounds.

2.2 Overall dimensions. Overall dimensions shall be the minimum consistent with the operational performance and the design constraints necessary to achieve the specified performance. Overall dimensions shall not exceed:

Length: 280 inches without couplers
Width: 120 inches without mirrors
Height: 100 inches

2.3 Drawbar pull (DBP). The tow tractor shall have a minimum DBP of 72,000 pounds on dry level concrete.

Comments, suggestions, or questions on this document should be addressed to WR-ALC/LGEC, 480 Richard Way Blvd. Suite 200, Robins AFB, GA 31098-1640 or emailed to bob.yohe@robins.af.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil

AMSC N/A FSC 1740

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2.4 Power train. The tow tractor shall have a commercially available, liquid cooled, turbocharged diesel engine of not less than six cylinders and power train sufficient to develop the required drawbar pull and obtain the required tow tractor speeds. Power train components shall have adequate torque and power ratings for the imposed torque and power. The tow tractor power train shall possess sufficient cooling capacity to perform continuous maximum performance towing in high ambient temperatures of +125 degrees F without exceeding the manufacturer's maximum allowable temperatures on the individual power train components. The diesel engine shall operate satisfactorily when using Jet A-1 (JP-8) fuel. Power train shall feature a standard (type C) winterization package that is good for starting down to minus 20F, or when specified (see 6.2 c), a heavy-duty (type A) winterization package good for storage down to minus 65F and starting to minus 40F. Winterization package components with external power requirements shall be compatible with 110VAC/20A circuitry. The tow tractor shall meet applicable 50-state exhaust and evaporative emissions standards in effect at the time of vehicle manufacture. Engine shall have an automatic engine shutdown system for high coolant temperature and low oil pressure. Transmission shall be of the power-shift type, with an integral, limited-slip, inter-axle differential. Maximum allowable transmission temperature during all operations shall not exceed 260 degrees F. Axles shall be planetary-drive/steer axles.

2.5 Driveline and steering. The tow tractor shall have four-wheel drive, and two-wheel and four-wheel power steering. Four-wheel steering shall feature both crab and coordinated steering. The selection of the steering mode shall be accomplished by the movement of a one-hand-operated lever or switch. When switching from the crab or coordinated four-wheel mode of steering to the two-wheel mode of steering, the rear wheels shall center automatically (wheels perpendicular to axle) as the steering wheel is rotated to the wheel center, without additional operator input or adjustment. The tow tractor shall have a maximum wall-to-wall turning radius of 26 feet or less while in four-wheel coordinated steering mode. Wheel-wells shall be designed so that tire chains may be used without interference.

2.6 Electro-magnetic interference (EMI). The tow tractor shall comply with EMI requirements of MIL-STD-461 for Air Force ground equipment.

2.7 Mobility. The tow tractor shall be capable of traveling a minimum of 15 miles per hour (MPH) forward and 2.5 MPH in reverse on flat asphalt or concrete surfaces without a trailing load.

2.8 Suspension. On occasion the tow tractor will leave the flight lines and roads within the perimeter of the base, and enter both paved and unpaved surfaces. The angle of each surface relative to the other can differ. The tow tractor shall have a suspension that will allow for articulation of one or both of the tow tractor's axles, so that an axle's position relative to the other will allow for four-wheel contact with the normal working surfaces found in the above described conditions.

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2.9 Wheels and tires. The tow tractor shall be equipped with steel disc wheels and tubeless steel belted radial tires with an appropriate tread pattern. The wheels and tires shall be in accordance with (IAW) the Tire and Rim Association recommendations for this application. All wheels and tires shall be uniform. Tire inflation pressures shall be noted on all wheel well openings. If tire loads and tire inflation pressures that exceed the specifications on the tire sidewall markings are required, a copy of the application approval from the tire manufacturer shall be added in the Technical Order.

2.10 Braking system. The tow tractor shall feature a four-wheel, hydraulically operated braking system with power assist. If an air system is used as the power assist, cab instrumentation shall include an air pressure gauge and a low-pressure indicator with buzzer and/or warning light (see 2.15).

2.11 Parking brake. The tow tractor shall be equipped with a spring applied and hydraulic release parking brake. A lockout system shall be provided so that the tow tractor will shut down when the parking brake is on and the tow tractor is placed in forward or reverse gear. The lockout system shall also include a cab-mounted warning light.

2.12 Air system. Reserved.

2.13 Electrical system. The tow tractor shall be equipped with a l2V or 24V electrical system. The battery(ies) shall be maintenance free.

2.14 Frame structure. The tow tractor shall have an all-welded, one-piece frame structure.

2.15 Cab design. The tow tractor shall be equipped with an enclosed two-man cab. Interior cab sound level shall not exceed 84 dB (A) with doors and windows closed. The first entrance step to the cab shall not exceed 18 inches in height from the ground. Cab instrumentation shall be standard commercial installation to include an hour meter, odometer, tachometer, speedometer, voltmeter, oil pressure gauge, coolant temperature gauge, and a fuel level gauge. If an air system is used as the power assist for the braking system (see 2.10), cab instrumentation shall include an air pressure gauge and a low-pressure indicator with buzzer and/or warning light. The tow tractor cab shall he equipped with wipers, driver and passenger-door rear-view mirrors, windshield-area rear-view mirror, visors, heater/defroster, ventilating fan, interior dome light, seat belts, and other equipment as found in standard commercial offerings. The seats shall incorporate a suspension device for driver and passenger comfort.

2.16 Lighting. The tow tractor lighting shall be equivalent to the standard commercial offering to include a rear coupling light, parking lights, head lights (high and low beam), brake lights, turn signals, and hazard lights. A flashing amber beacon light shall be mounted on the cab roof.

2.17 Couplers. The tow tractor shall be equipped with front and rear clevis-pin type couplers. The clevis-pin type couplers shall be able to accommodate 3-inch ID lunettes (drawbar ring). The couplers shall be located such that the coupling positions are at 20 inches above the ground.

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2.18 Breakaway cable hookup. Reserved.

2.19 Safety chain attachment points. Reserved.

2.20 Trailer light harness receptacle. Reserved.

2.21 Fire extinguisher. The tow tractor shall be equipped with a 17 pound (minimum) dry chemical type fire extinguishers with mounting brackets, and a ULC rating of 20B:C (minimum). The fire extinguisher shall be mounted on the rear side of the tow tractor.

2.22 Finish. As specified (see 6.2 d), the tow tractor shall be painted Dark Green (Color 24052 per FED-STD-595) or Desert Tan (Color 23448 per FED-STD-595) using offeror's standard commercial paint system.

2.23 Corrosion protection. The tow tractor shall be rust proofed and undercoated IAW FED-STD-297. The level of corrosion treatment shall be tropical, and the use of proprietary material shall require Government approval.

2.24 Visibility. The tow tractor shall be so constructed as to provide maximum visibility in all directions. Rearward visibility shall be provided within an envelope defined by a point one foot behind the tractor and 54 inches above the ground to a point fifteen feet from the rear of the tractor and 30 inches from the ground. Additionally, the front and rear clevis-pin type couplers shall be visible from the driver's position either directly or via a single mirror mounted on the front and rear of the tow tractor.

2.25 Air transportability. The tow tractor shall be air transportable on C-17 and C-5 aircraft. Use MIL-HDBK-1791 as a guide. In all transport configurations, the tow tractor shall be capable of being restrained and withstanding 2.0 G up, 4.5 G down, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The tow tractor shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure.

2.25.1 Air transportable configuration. The air transportable configuration shall include all equipment, ¾ tank of fuel, and no crewmembers. In this configuration, maximum weight shall be 72,000 pounds, with a maximum single axle weight of 36,000 pounds. At least 6.0 inches sidewall and 6.0 inches overhead clearance shall be maintained between the tow tractor and the aircraft at all times during loading and flight. The restrained tow tractor shall allow for loadmaster in-flight access from the front to the rear of the aircraft.

2.25.2 Equipment removal/reconfiguration. Preparation and restoration of the tow tractor for air transportability shall take no more than 30 minutes for two persons using common non-powered hand tools. All equipment removed shall be stored on the tow tractor; caps and plugs shall permit driving and storage in transport configuration.

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2.25.3 Tie down points. The tow tractor shall be symmetrically restrained during air transport. Tie down points shall be rated at a minimum of 25,000 pounds, marked for capacity, with a clear opening compatible with the appropriate devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the tow tractor.

2.25.4 Shoring. The use of shoring to load the aircraft (referred to as approach shoring and step-up shoring) and during flight (referred to as load shoring, parking shoring, and sleeper shoring) is permitted, but not desired. However, the use of shoring, especially in loading the aircraft, is not an alternative to good tow tractor design. The tow tractor shall be designed to the maximum extent possible with adequate clearances to load onto the aircraft.

2.25.5 Removable Ballast. The use of removable ballast is permitted to bring the tow tractor down to its maximum air transportable weight. Any one piece of ballast weight shall not weigh more than 8,500 pounds, shall be of a size that permits it to be removed with a 10 K pound forklift with 8.5" wide holes for 8" wide tines, shall be loadable onto one HCU-6/E 463L pallet with maximum size 84" L x 104" W, and shall have tie down provisions in sufficient number and located so as to be capable of being restrained to the pallet to a minimum of 3.0 G in the plane of the pallet and 2.0 G normal to the pallet. The ballast shall present a load weight onto the pallet surface (with shoring if necessary) that does not exceed 100 pounds/linear inch (C-5 roller limit) or 250 psi pallet contact pressure. If removable ballast weight is incorporated to fulfill this requirement, ballasts shall have their respective weights stencil painted onto each removable piece.

2.26 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information.

1. Side and rear silhouette views of the tow tractor
2. Horizontal and vertical location of the center of gravity of the tow tractor in air transportable configuration, marked on the silhouette views
3. Shipping weight
4. Loading cubage
5. Overall height, width, and length
6. Front and rear axle loads
7. Tie down information

2.27 Identification plate. The tow tractor shall have an identification plate and a rear license plate mounting bracket. The identification plate shall include as a minimum the following.

1. Vehicle Nomenclature
2. Manufacturer’s Model Number
3. Vehicle Identification (serial) Number or VIN
4. Manufacturer’s address and phone number
5. Date of Manufacture
6. National Stock Number
7. Contract Number
8. Registration Number

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2.28 Lubrication data plate. The tow tractor shall be provided with a lubrication data plate that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

2.29 Miscellaneous specifications. The tow tractor shall comply with Society of Automotive Engineers (SAE) specifications ARP1247 and AIR 1363.

2.30 Technical Order (TO) requirements. The contractor shall provide copies of the tow tractor manuals for review 30 days prior to the time of tests or demonstration of the first tow tractor. The contractor's parts manuals shall list the part numbers for contractor-manufactured parts, plus the part numbers for purchased sub-level components and whole components. The contractor shall also provide with each tow tractor, a product familiarization videotape that verbally and visually provides all information for the operation and routine maintenance of the tow tractor and its components, using the commercial manuals as a baseline.

3. RECLAIMED MATERIAL. The contractor shall utilize reclaimed materials to the maximum extent practical. Used, rebuilt, or remanufactured components shall not be incorporated in the tow tractor.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be substantially the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Commercial item requirement. The tow tractor furnished must meet the "commercial item" definition, as specified in FAR 2.101, as of the date of award. The offeror/contractor shall provide two copies of their commercial, descriptive catalogs with their offer, as specified in Clause I106. The offeror/contractor shall identify all modifications made to their commercial model in order to meet the performance and descriptive requirements of the CID or the referenced documents. In regard to the offered item, the offeror/contractor shall identify any enhancements or improvements to the performance requirements of paragraph 2 of the CID or the referenced documents.

4.3 Verification. Prior to delivery of the first tow tractor, the contractor shall demonstrate or test at his facility, each requirement of the CID or referenced documents.

4.3.1 Gross vehicle weight (GVW). The first production tow tractor shall be weighed for verification of the minimum GVW.

4.3.2 Overall dimensions. The first production tow tractor shall be measured for verification of the overall dimensions.

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4.3.3 Drawbar pull (DBP). The first production tow tractor shall be tested for verification of the DBP on dry level concrete using a load sensing device and a fixed anchoring position. The drawbar position shall be horizontal and level.

4.3.4 Power train. The first production tow tractor shall be inspected for verification of the specified power train components. A copy of the manufacturer's signed application for approval shall be submitted to the government representative at first production verification with the vehicle documents for the following components: engine, transmission/transfer case, and axles, to include torque and power ratings, use of Jet A-1 or JP-8 fuels, emissions compliance, and engine shutdown system. The cooling system shall be certified by the vehicle manufacturer to satisfy all operating conditions at all ambient temperatures up to the specified, for both the engine and transmission. The winterization packages shall be certified by the vehicle manufacturer to satisfy the winterization conditions specified.

4.3.5 Driveline and steering. The first production tow tractor shall be demonstrated in two-wheel steer and four-wheel steer, using both crab and coordinated steering. The tow tractor shall be tested for verification of tire chain use without interference in wheel-wells. Tire chains shall be fitted to the tires, and the tow tractor shall be operated by driving a short distance and turning the wheels to the extreme left and right. The tow tractor shall be tested for verification of the maximum wall-to-wall turning radius while in four-wheel coordinated steering mode. The fully ballasted first production tow tractor shall be tested as follows to demonstrate compliance with the turning radius specification:

a. Facility. Perform the test on a dry, level, paved area that is free from loose material and is larger in all directions than three times the length of the vehicle being tested.
b. Equipment required.
    1. A device suitable for measuring three times the length of the vehicle being tested with an accuracy of at least ±1-inch.
    2. Markers or marking device suitable for marking the pavement.
    3. A plumb bob suitable for locating a point on the pavement directly below a fixed point on the vehicle.
    4. A test report notebook or similar record forms to be used as a test report work sheet and incorporated into the documentation package.
c. Test Conditions.
    1. The tow tractor shall be fully ballasted and in operational form.
    2. The tow tractor steering system shall be fully operational, and the steering linkage stops shall be adjusted to the manufacturer's specified production tolerance limits.
d. Test Procedure.
    1. The vehicle shall be driven slowly in a full cramp circle (left or right) to establish a steady state in the steering linkage.
    2. Continue driving the slow full cramp circle.
    3. At approximately three equidistant points (identified as A, B, and C) around the circle, gently stop the vehicle using the service brakes.
    4. At each stop, place a plumb bob against the outermost point of the vehicle and mark the spot on the ground directly below where the plumb bob comes to rest.

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5. Measure and record the straight line distances between each pair of points, i.e., Lengths AB, BC, and CA.
6. Calculate the wall-to-wall turning radius (R) as follows:

   R = D / 2 where

   D = (AB x BC x CA) / (2 x v(S x (S-AB) x (S-BC) x (S-CA))) and

   S = (AB + BC + CA) / 2

7. Repeat steps (1) through (6) with the vehicle moving in the opposite direction.

4.3.6 Electro-magnetic interference (EMI). The first production tow tractor shall be tested by the contractor for verification of the specified EMI requirements prior to first production verification of the tow tractor. The tow tractor shall be tested for Radiated Emission per RE102 of MIL-STD-461 and Radiated Susceptibility per RS103 of MIL-STD-461. The tow tractor shall be tested for Conducted Susceptibility per CS114, CS115, and CS116 of MIL-STD-461. Test results shall be submitted to the government representative at first production verification.

4.3.7 Mobility. The first production tow tractor shall be tested for verification of the forward and reverse speeds on a flat asphalt or concrete surface without a trailing load, using the tow tractor's speedometer as the speed-measuring device.

4.3.8 Suspension. The first production tow tractor shall be inspected for verification of a suspension that will allow for articulation of the tow tractor's axles relative to each other.

4.3.9 Wheels and tires. The first production tow tractor shall be inspected for verification of the specified wheels and tires. Documentation of the proper wheel and tire selection and deviations in tire pressures (if any) with references to the Tire and Rim Association shall be submitted to the government representative at the first production verification.

4.3.10 Braking system. The first production tow tractor shall be inspected for verification of a four-wheel, hydraulically operated braking system with power assist. A schematic of the braking system shall be submitted to the government representative at the first production verification.

4.3.11 Parking brake. The first production tow tractor shall be inspected for verification of the spring applied and hydraulic release parking brake, and the parking brake lockout system.

4.3.12 Air system. Reserved.

4.3.13 Electrical system. The first production tow tractor shall be inspected for verification of the specified electrical system.

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4.3.14 Frame structure. The first production tow tractor shall be inspected for verification of the specified all-welded, one-piece frame structure.

4.3.15 Cab design. The first production tow tractor shall be inspected for verification of the enclosed two-man cab, the entrance step height, cab instrumentation, and specified equipment. The tow tractor shall be tested for verification of the interior sound level using a sound level meter at the operator and passenger ear level, with the doors and windows closed, and the vehicle and accessories running at engine idle, mid-range power, and full power.

4.3.16 Lighting. The first production tow tractor shall be inspected for verification of the specified lighting.

4.3.17 Couplers. The first production tow tractor shall be inspected for verification of the specified couplers and locations.

4.3.18 Breakaway cable hookup. Reserved.

4.3.19 Safety chain attachment points. Reserved.

4.3.20 Trailer light harness receptacle. Reserved.

4.3.21 Fire extinguisher. The first production tow tractor shall be inspected for verification of the specified fire extinguishers and mounting brackets.

4.3.22 Finish. Color sample cards with the specified colors shall be submitted at the first production verification.

4.3.23 Corrosion protection. The first production tow tractor shall be inspected for verification of the specified rust proofing and undercoating. Documentation of the application process and certification of the level of corrosion treatment shall be submitted to the government representative at the first production verification.

4.3.24 Visibility. The first production tow tractor shall be inspected for verification of the specified visibility. Measurements of the two points behind the tow tractor shall be made, and verification of the visibility of those two points shall be performed.

4.3.25 Air transportability. An air transportability report shall be submitted to the government representative at first production verification to begin the air transportability certification process. The report shall outline the steps necessary to load the tow tractor onto the specified aircraft, including diagrams, drawings, or instructions of the tow tractor preparation, loading, shoring, and tie down procedures (including engineering analysis of the tie down devices) for both the tow tractor and any removable ballasts, if applicable, along with critical dimensions, tow tractor weights, axle weights, etc.

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4.3.26 Transportation data plate. The first production tow tractor shall be inspected for verification of the specified transportation data plate.

4.3.27 Identification plate. The first production tow tractor shall be inspected for verification of the specified identification plate and a rear license plate mounting bracket.

4.3.28 Lubrication data plate. The first production tow tractor shall be inspected for verification of the specified lubrication data plate.

4.3.29 Miscellaneous specifications. The first production tow tractor shall be inspected for verification of compliance with SAE specifications ARP1247 and AIR 1363.

4.4 Reconfiguration criteria. Failure to comply with the requirements of this CID, reference documents, or the tests shall be cause for reconfiguration and re-demonstration. Causes for reconfiguration shall include: structural failure; non-correctable misalignments; component interference; conditions presenting a safety hazard to the user or maintainer: instability during operation, spillage of fuel or coolant, and overheating.

4.5 Operational tests. If specified in the ordering data (see 6.2 e), contractor shall provide a specified number of sample units to be placed in service at representative Air Force bases. These sample units shall be used interchangeably with existing U-30s to determine their fitness for use. The period of testing and evaluation shall be 90 days.

5. PACKAGING.

5.1 Preservation, packing, and marking shall be as specified in the contract or order (see 6.2 b).

6. NOTES.

6.1 Addresses for obtaining copies of referenced documents.

Copies of Department of Defense specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or www.dodssp.daps.mil.

Application for copies of SAE documents should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or www.sae.org/.

6.2 Ordering data. The purchaser must specify the following at the time of purchase:

a. Title, number, and revision letter, and date of this document.
b. Packaging requirements (see 5.1).
c. When heavy-duty winterization is required (see 2.4).
d. Color required (see 2.22).
e. Operational tests if required (see 4.5).

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6.3 Subject term (key word) listing.

Aircraft parking
Ground handling
Tug

Custodians:
Air Force – 99
Army – AV

Review activity:
Army – MI

Preparing activity:
Air Force – 84

Agent
Air Force – 99

Civilian coordinating activity:
GSA/FSS

(Project 1740-0193)

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